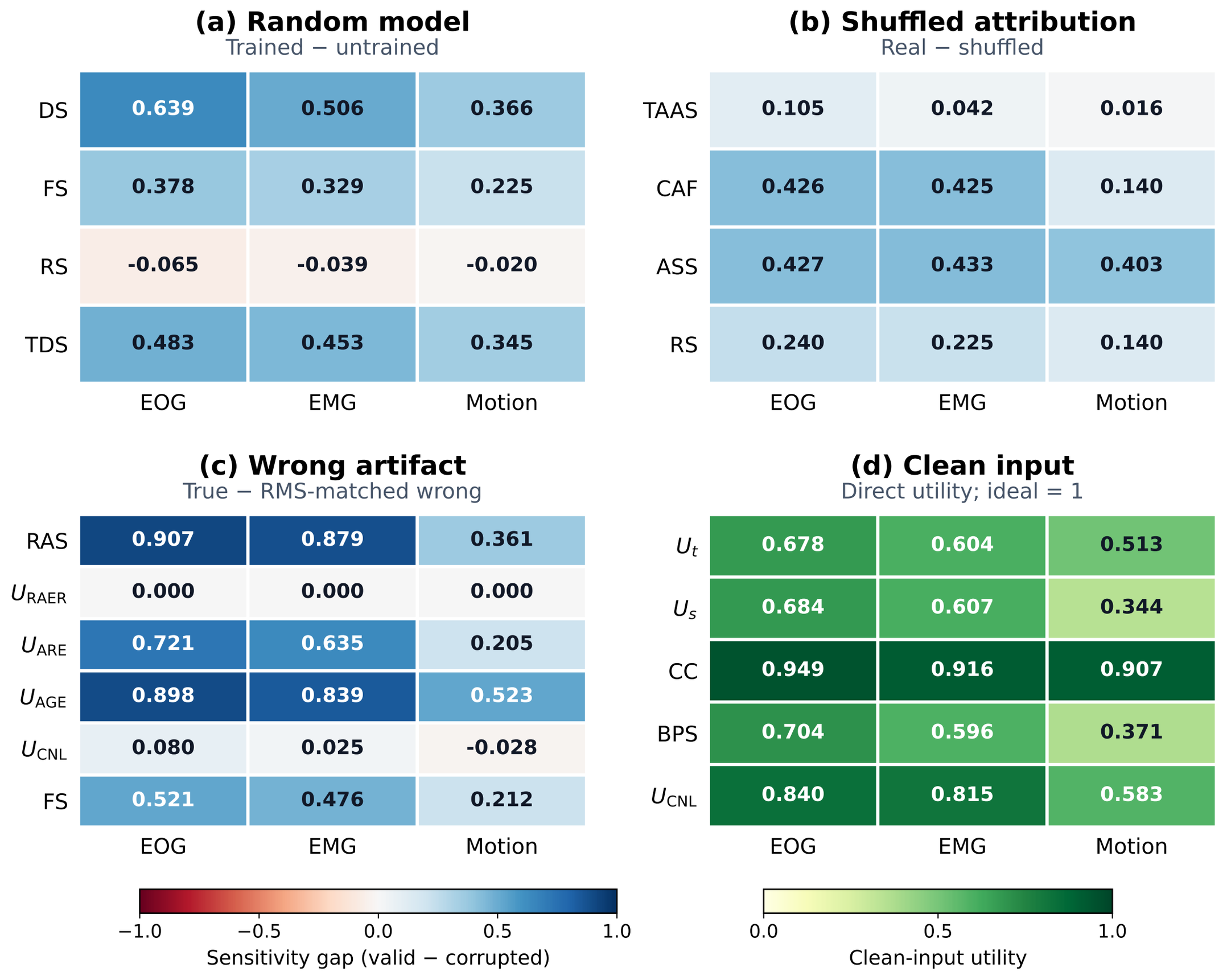
(a) Random model
Trained − untrained
DS
FS
RS
TDS
0.639
0.506
0.366
0.378
0.329
0.225
-0.065
-0.039
-0.020
0.483
0.453
0.345
EOG
EMG
Motion
(b) Shuffled attribution
Real − shuffled
TAAS
CAF
ASS
RS
0.105
0.042
0.016
0.426
0.425
0.140
0.427
0.433
0.403
0.240
0.225
0.140
EOG
EMG
Motion
(c) Wrong artifact
True − RMS-matched wrong
RAS
$U_{RAER}$
$U_{ARE}$
$U_{AGE}$
$U_{CNL}$
FS
0.907
0.879
0.361
0.000
0.000
0.000
0.721
0.635
0.205
0.898
0.839
0.523
0.080
0.025
-0.028
0.521
0.476
0.212
EOG
EMG
Motion
−1.0
−0.5
0.0
0.5
1.0
Sensitivity gap (valid − corrupted)
(d) Clean input
Direct utility; ideal = 1
$U_t$
$U_s$
CC
BPS
$U_{CNL}$
0.678
0.604
0.513
0.684
0.607
0.344
0.949
0.916
0.907
0.704
0.596
0.371
0.840
0.815
0.583
EOG
EMG
Motion
0.0
0.5
1.0
Clean-input utility